

Lesson 20: Bootstrap and Permutation Methods for Two Samples

BSTA 551: Statistical Inference

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Review: Lessons 18–19

Parametric two-sample methods so far:

- **Lesson 18:** Welch t / pooled t for $\mu_1 - \mu_2$; z -test for $p_1 - p_2$; F -test for σ_1^2 / σ_2^2
- **Lesson 19:** Paired t -test; variance reduction from positive within-pair correlation

All require **normality** (exact) or rely on the **CLT** (approximate). What if:

- Samples are small and distributions are visibly skewed?
- The parameter of interest is a **median, IQR, ratio**, or other non-mean statistic?
- Distributional assumptions are questionable?

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- Samples are small and distributions are visibly skewed?
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Today's Goals:

1. Construct **two-sample bootstrap CIs** by resampling each group independently (C&H Ch. 3; DBC 10.6)
2. Implement the **two-sample permutation test** by permuting group labels (C&H Ch. 5)
3. Apply bootstrap to **paired data** by resampling whole pairs
4. Articulate the precise conceptual difference between bootstrap and permutation
5. Compare resampling to parametric methods through simulation

Roadmap

Method	Goal	Resamples From	Reference
Two-sample bootstrap CI	CI for $\theta_1 - \theta_2$	Each group separately	C&H Ch. 3; DBC 10.6
Two-sample permutation test	Test $H_0 : F_1 = F_2$	Pooled, relabeling	C&H Ch. 5
Paired bootstrap CI	CI for μ_D	Pairs with replacement	C&H Ch. 3

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Paired bootstrap CI	CI for μ_D	Pairs with replacement	C&H Ch. 3

Motivating Example

Trauma registry: Compare hospital length of stay (LOS, days) between blunt and penetrating trauma patients. LOS is strongly right-skewed and n is moderate: the Welch t -test CI may be unreliable. Bootstrap and permutation give valid inference with no distributional assumptions.

Two-Sample Bootstrap CI

The Two-Sample Bootstrap Algorithm (C&H 3.2)

Setup: Independent samples $x_1, \dots, x_{n_1}$ from F_1 and $y_1, \dots, y_{n_2}$ from F_2 .

Goal: CI for $\theta = g(F_1, F_2)$, e.g., $\mu_1 - \mu_2$, $\text{median}_1 - \text{median}_2$, σ_1/σ_2 .

Algorithm: for $b = 1, \dots, B$:

1. Draw $x_1^{*b}, \dots, x_{n_1}^{*b}$ **with replacement** from $\{x_1, \dots, x_{n_1}\}$
2. Draw $y_1^{*b}, \dots, y_{n_2}^{*b}$ **with replacement** from $\{y_1, \dots, y_{n_2}\}$
3. Compute $\hat{\theta}^{*b} = g(\mathbf{x}^{*b}, \mathbf{y}^{*b})$

Percentile CI: $(\hat{\theta}_{(\alpha/2)}^*, \hat{\theta}_{(1-\alpha/2)}^*)$

Basic (reflection) CI: $(2\hat{\theta} - \hat{\theta}_{(1-\alpha/2)}^*, 2\hat{\theta} - \hat{\theta}_{(\alpha/2)}^*)$

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! Key Point

Resample each group **separately and independently** — this preserves the two-group structure and approximates the sampling distribution of $\hat{\theta}_1 - \hat{\theta}_2$. Pooling before resampling would impose H_0 (that is permutation, not bootstrap).

Medical Example: Trauma Length of Stay

► Code

Blunt: n = 40 mean = 7.46 median = 6.29 sd = 5.26

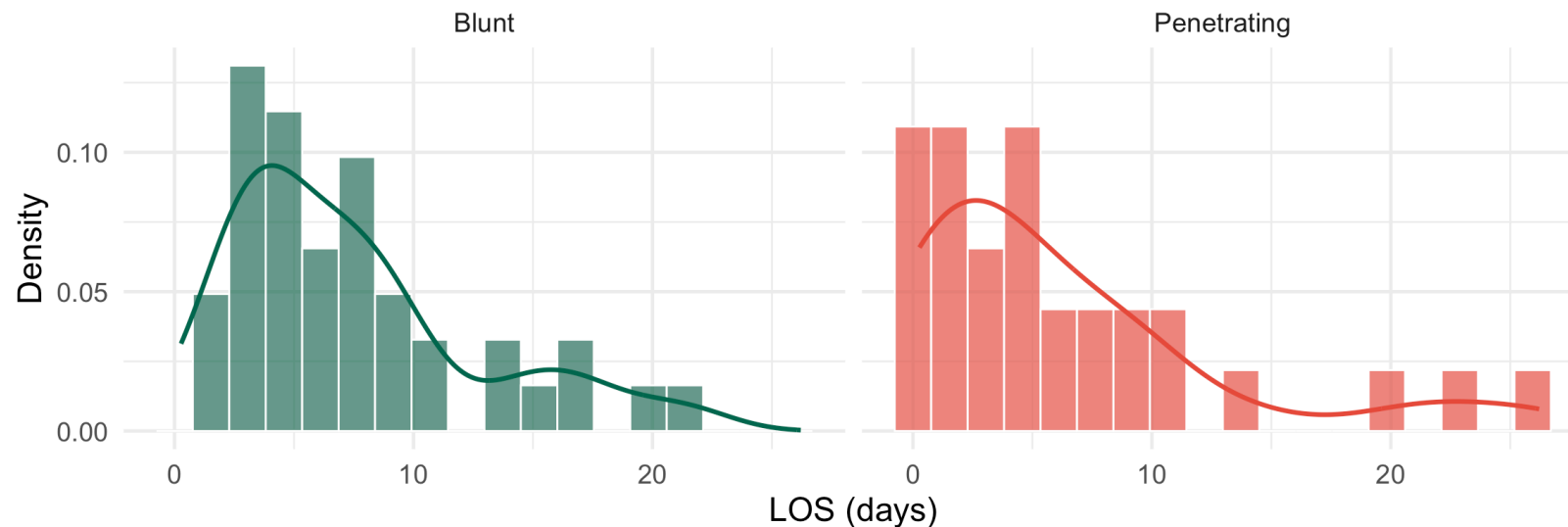
► Code

Penetrating: n = 30 mean = 6.39 median = 4.28 sd = 6.76

► Code

Hospital Length of Stay by Trauma Type

Right-skewed distributions — parametric CIs may be unreliable for moderate n



Computing the Bootstrap CI

```
1 set.seed(551)
2 B      <- 5000
3 n1     <- length(los_blunt)
4 n2     <- length(los_penet)
5 obs_diff <- mean(los_blunt) - mean(los_penet)
6
7 boot_diffs <- replicate(B, {
8   mean(sample(los_blunt, n1, replace = TRUE)) -
9   mean(sample(los_penet, n2, replace = TRUE))
10 })
11
12 ci_pct <- quantile(boot_diffs, c(0.025, 0.975))
13 ci_basic <- 2 * obs_diff - quantile(boot_diffs, c(0.975, 0.025))
14 ci_welch <- t.test(los_blunt, los_penet)$conf.int
15
16 cat(sprintf("Observed difference (Blunt - Penet):  %.3f days\n", obs_diff))
```

Observed difference (Blunt - Penet): 1.070 days

```
1 cat(sprintf("Bootstrap SE:                               %.3f\n", sd(boot_diffs)))
```

Bootstrap SE: 1.491

```
1 cat(sprintf("Percentile 95% CI:  (%.3f, %.3f)\n", ci_pct[1], ci_pct[2]))
```

Percentile 95% CI: (-1.979, 3.868)

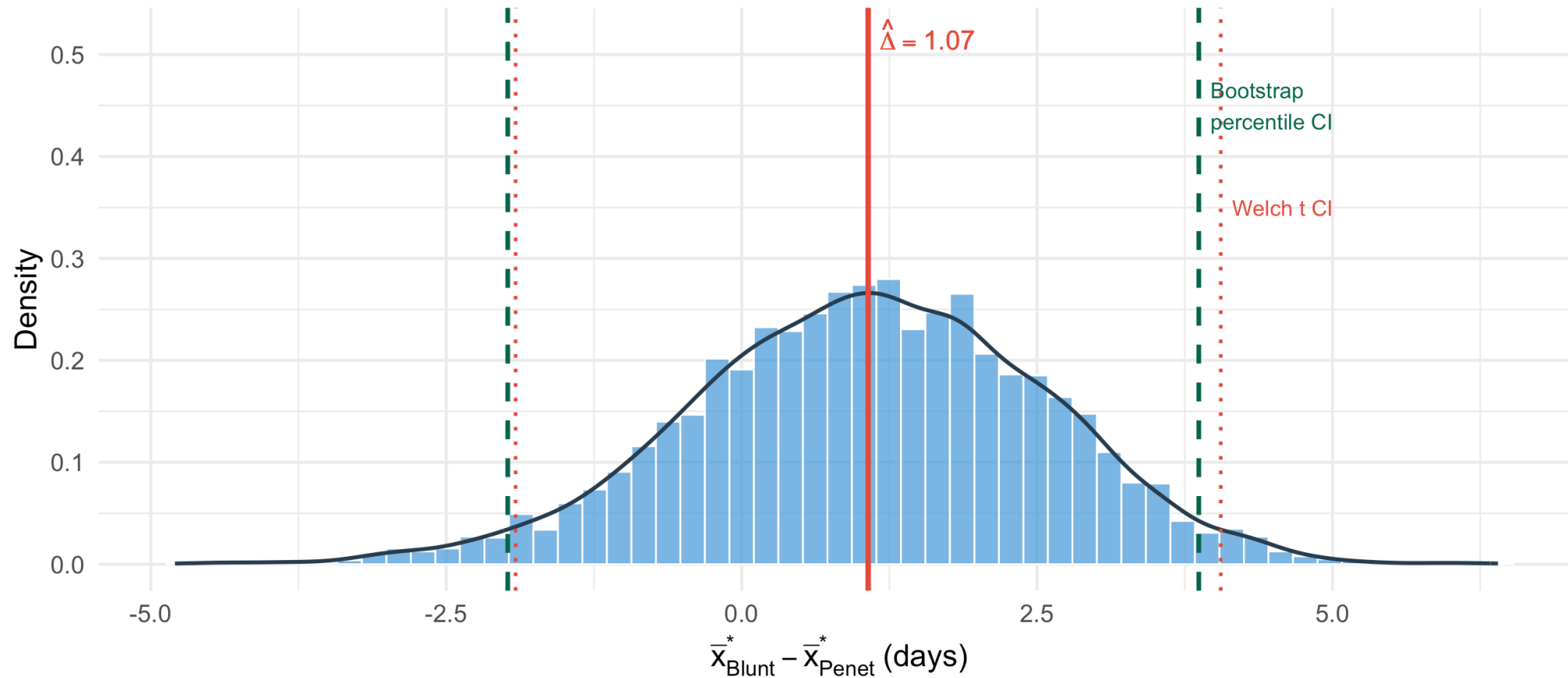


Visualizing the Bootstrap Distribution

► Code

Bootstrap Distribution: Mean LOS Difference (Blunt – Penetrating)

Green dashed = 95% bootstrap percentile CI | Red dotted = Welch t CI



Bootstrap CI for Any Statistic

A key advantage over parametric methods: the same algorithm works for **any statistic**.

```
1 set.seed(551)
2
3 # Difference in medians (no closed-form parametric CI exists)
4 boot_med <- replicate(B, {
5   median(sample(los_blunt, n1, replace = TRUE)) -
6   median(sample(los_penet, n2, replace = TRUE))
7 })
8
9 # Ratio of standard deviations
10 boot_sdr <- replicate(B, {
11   sd(sample(los_blunt, n1, replace = TRUE)) /
12   sd(sample(los_penet, n2, replace = TRUE))
13 })
14
15 cat("--- Difference in MEDIANs ---\n")
```

--- Difference in MEDIANs ---

```
1 cat(sprintf("  Observed: %.3f days\n",
2             median(los_blunt) - median(los_penet)))
```

Observed: 2.003 days

```
1 cat(sprintf("  Bootstrap 95% CI: (%.3f, %.3f)\n\n",
```

Two-Sample Permutation Test

The Permutation Test: Core Idea (C&H Ch. 5)

Null hypothesis: $H_0 : F_1 = F_2$ — the two groups have the **same distribution**.

Under H_0 , the group label on each observation is **arbitrary** (exchangeable). Any assignment of the $N = n_1 + n_2$ pooled observations into groups of size n_1 and n_2 is equally likely.

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Algorithm: for $b = 1, \dots, B$:

1. Pool all N observations
2. Randomly permute labels: assign the first n_1 to group 1, the rest to group 2
3. Compute T^{*b} (the test statistic under the permuted labeling)

Two-sided p-value: $\hat{p} = \frac{1}{B} \sum_{b=1}^B \mathbf{1}(|T^{*b}| \geq |T_{\text{obs}}|)$

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⚠ Bootstrap vs. Permutation – The Essential Distinction

	Bootstrap CI	Permutation test
Purpose	Estimate variability	Test H_0
Resamples from	Each group separately	Pooled data (relabeling)
Approximates	Sampling distribution of $\hat{\theta}$	Null distribution of T
H_0 imposed?	No	Yes ($F_1 = F_2$)

Permutation Test: Implementation

```
1 set.seed(551)
2 B_perm <- 9999
3 all_los <- c(los_blunt, los_penet)
4 N <- length(all_los)
5 t_obs <- mean(los_blunt) - mean(los_penet)
6
7 perm_diffs <- replicate(B_perm, {
8   idx <- sample(N)
9   mean(all_los[idx[seq_len(n1)]] - mean(all_los[idx[(n1 + 1):N]]))
10 })
11
12 p_perm <- mean(abs(perm_diffs) >= abs(t_obs))
13 p_welch <- t.test(los_blunt, los_penet)$p.value
14
15 cat(sprintf("Observed mean difference:   %.3f days\n", t_obs))
```

Observed mean difference: 1.070 days

```
1 cat(sprintf("Permutation p-value:      %.4f\n",      p_perm))
```

Permutation p-value: 0.4551

```
1 cat(sprintf("Welch t-test p-value:      %.4f\n",      p_welch))
```

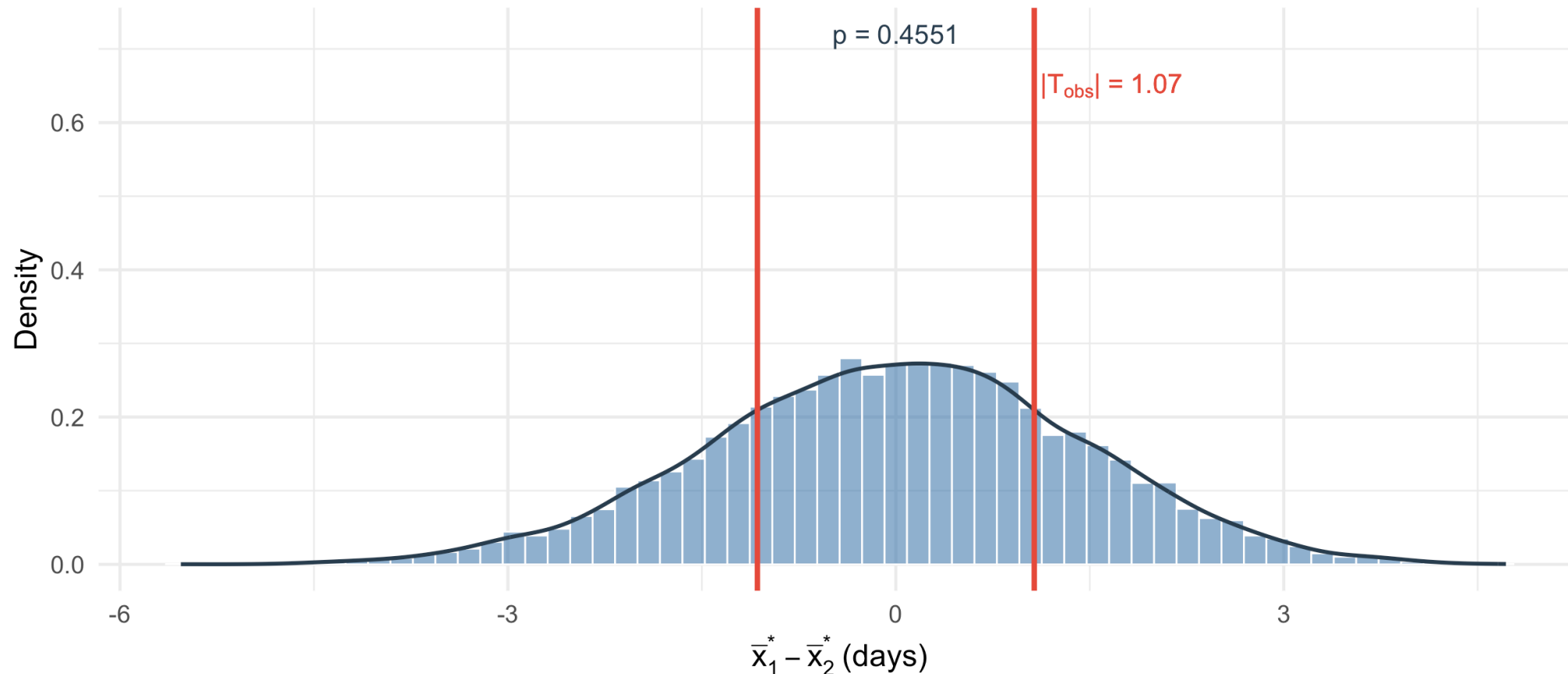
Welch t-test p-value: 0.4749

Visualizing the Permutation Null Distribution

► Code

Permutation Null Distribution: Difference in Mean LOS

Red lines mark $\pm|T_{\text{obs}}|$. Fraction of permuted statistics beyond = p-value.



Permutation Tests Work with Any Statistic

```
1 set.seed(551)
2
3 # Permutation test for difference in MEDIANs
4 t_obs_med <- median(los_blunt) - median(los_penet)
5 perm_med  <- replicate(9999, {
6   idx <- sample(N)
7   median(all_los[idx[seq_len(n1)]] - median(all_los[idx[(n1 + 1):N]]))
8 })
9 p_perm_med <- mean(abs(perm_med) >= abs(t_obs_med))
10
11 cat(sprintf("Median difference: %.3f    permutation p = %.4f\n",
12            t_obs_med, p_perm_med))
```

Median difference: 2.003 permutation p = 0.2916

```
1 # Permutation test for VARIANCE RATIO (vs F-test)
2 t_obs_var <- var(los_blunt) / var(los_penet)
3 perm_var  <- replicate(9999, {
4   idx <- sample(N)
5   var(all_los[idx[seq_len(n1)]] / var(all_los[idx[(n1 + 1):N]]))
6 })
7 p_perm_var <- mean(perm_var >= t_obs_var | perm_var <= 1 / t_obs_var)
8 p_ftest    <- var.test(los_blunt, los_penet)$p.value
9
```



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8 p_ftest    <- var.test(los_blunt, los_penet)$p.value
9
```



Bootstrap for Paired Data

Paired Bootstrap CI

For paired data, the bootstrap resamples **entire pairs** (D_i) with replacement which preserves the pairing structure.

Algorithm: for $b = 1, \dots, B$:

1. Draw $D_1^{*b}, \dots, D_n^{*b}$ **with replacement** from $\{D_1, \dots, D_n\}$
2. Compute $\bar{D}^{*b} = \frac{1}{n} \sum D_i^{*b}$

Percentile CI: $(\bar{D}_{(\alpha/2)}^*, \bar{D}_{(1-\alpha/2)}^*)$

No normality assumption on D_i .

```
1 # Crossover trial differences from Lesson 19
2 set.seed(42)
3 n_pair <- 20; rho_p <- 0.75
4 L_p <- chol(10^2 * matrix(c(1, rho_p, rho_p, 1), 2))
5 xy_p <- matrix(rnorm(2 * n_pair), n_pair, 2) %*% L_p +
6         matrix(c(0, -10), n_pair, 2, byrow = TRUE)
7 d_cross <- xy_p[, 2] - xy_p[, 1]
8
9 set.seed(551)
10 boot_D <- replicate(5000, mean(sample(d_cross, n_pair, replace = TRUE)))
11 ci_boot_paired <- quantile(boot_D, c(0.025, 0.975))
```



Comparing Methods: Simulation

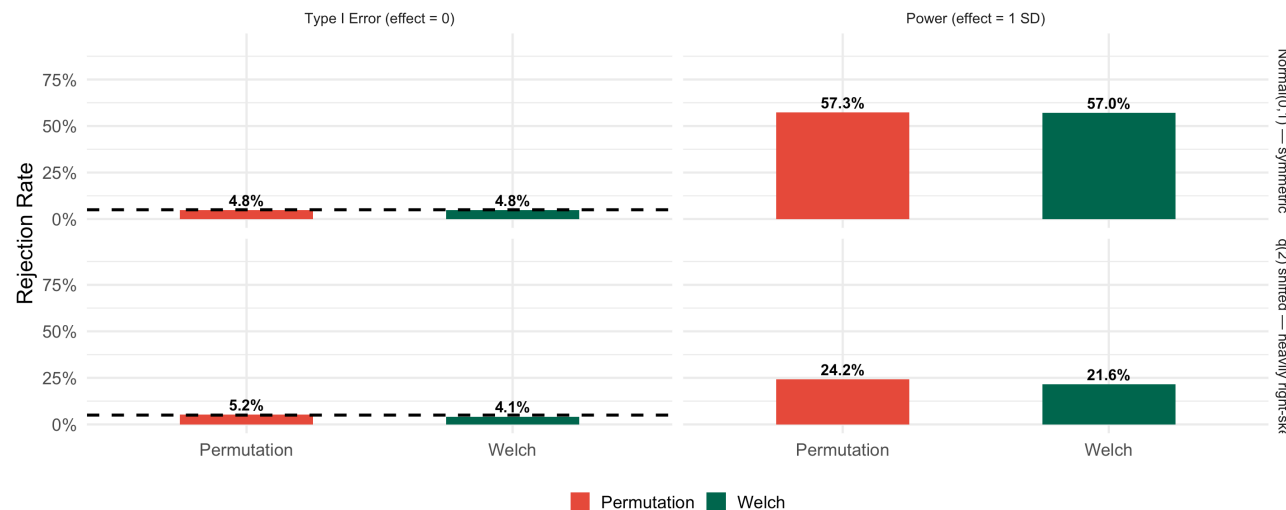
Type I Error and Power: Welch t-test vs. Permutation Test

The permutation test derives its null distribution empirically rather than from the t -distribution. For the **mean difference** statistic, how much does it matter?

► Code

Welch t-test vs. Permutation Test (n = 10 per group, 3000 simulations)

Both methods control Type I error near 5% and have similar power — permutation's advantage is flexibility.



i Note

The two-sided Welch t -test is **robust to skewness for Type I error** even at small n : skewness makes each tail individually wrong, but in a two-sided test the errors in the two tails largely cancel. The simulation below confirms this — the two methods agree closely. Permutation's advantage is **generality**: it works with *any* test statistic, not just means.

Key Takeaways

Key Takeaways (C&H Ch. 3, 5; DBC 10.6)

! Bootstrap and Permutation for Two Samples

1. **Two-sample bootstrap CI:** Resample each group **independently** with replacement; construct percentile or basic CI. Works for any statistic — means, medians, SD ratios, etc. No distributional assumption
2. **Permutation test:** Pool all N observations, **relabel** groups at random; fraction of permuted statistics $\geq |T_{\text{obs}}|$ gives a valid p-value under $H_0 : F_1 = F_2$ (exchangeability)
3. **Bootstrap $\neq$ permutation:** Bootstrap approximates the **sampling distribution** of $\hat{\theta}$ (for CIs); permutation approximates the **null distribution** of T (for testing); they solve different problems and use data differently
4. **Paired bootstrap:** Resample **whole pairs** (D_i) to preserve the pairing structure; produces CI for μ_D without normality assumption
5. **When to prefer resampling:** Small n with non-normal data; non-standard statistics with no closed-form SE; any setting where parametric assumptions are doubtful

Looking Ahead

Next class (Lesson 21): Nonparametric Inference (DBC Ch. 14)

- Sign test: based on the binomial distribution, zero distributional assumptions
- Wilcoxon signed-rank test: uses ranks of $|D_i|$, requires symmetric differences
- Wilcoxon rank-sum test (Mann–Whitney): independent samples, ranks all N observations
- Asymptotic relative efficiency vs. parametric counterparts

References

- Chihara, L. M. and Hesterberg, T. C. *Mathematical Statistics with Resampling and R* (Wiley). Chapters 3 and 5.
- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Section 10.6.

Questions?

Thank you!

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